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COURSE:CSA1718_ai

Experiment 9: Travelling Salesman Problem (TSP)

Aim

To write a Python program to solve the **Travelling Salesman Problem (TSP)** and find the minimum cost path that visits every city exactly once and returns to the starting city.

Procedure

1. Define the distance matrix between cities.
2. Generate all possible paths.
3. Calculate the total distance for each path.
4. Select the path with the minimum distance.
5. Display the shortest path and its cost.

Python Code

```
from itertools import permutations
```

```
graph = [  
    [0, 10, 15, 20],  
    [10, 0, 35, 25],  
    [15, 35, 0, 30],  
    [20, 25, 30, 0]  
]
```

```
cities = [0, 1, 2, 3]
```

```
min_cost = float('inf')
```

```
best_path = []
```

```

for path in permutations(cities[1:]):

    current = [0] + list(path) + [0]

    cost = 0

    for i in range(len(current)-1):

        cost += graph[current[i]][current[i+1]]

    if cost < min_cost:

        min_cost = cost

        best_path = current

print("Shortest Path:", best_path)

print("Minimum Cost:", min_cost)

```

Output

Shortest Path: [0, 1, 3, 2, 0]

Minimum Cost: 80

```

Python3
1 from itertools import permutations
2
3 graph = [
4     [0, 10, 15, 20],
5     [10, 0, 35, 25],
6     [15, 35, 0, 30],
7     [20, 25, 30, 0]
8 ]
9
10 cities = [0, 1, 2, 3]
11 min_cost = float('inf')
12 best_path = []
13
14 for path in permutations(cities[1:]):
15     current = [0] + list(path) + [0]
16     cost = 0
17
18     for i in range(len(current)-1):
19         cost += graph[current[i]][current[i+1]]
20
21     if cost < min_cost:
22         min_cost = cost
23         best_path = current
24
25 print("Shortest Path:", best_path)
26 print("Minimum Cost:", min_cost)

```

Run Visualize Code

Enter Input here

If your code takes input, add it in the above box before running

Output

Status : Successfully executed

Time:	Memory:
0.0200 secs	9.072 Mb

Your Output

```

Shortest Path: [0, 1, 3, 2, 0]
Minimum Cost: 80

```

Result

Thus, the Travelling Salesman Problem was successfully implemented, and the minimum travel cost was found.

Experiment 10: A* Algorithm

Aim

To write a Python program to implement the **A* Search Algorithm** for finding the shortest path between the start node and the goal node.

Procedure

1. Create a graph.
2. Assign heuristic values to each node.
3. Start from the source node.
4. Select the node with the lowest evaluation function $f(n)=g(n)+h(n)$.
5. Continue until the goal node is reached.
6. Display the shortest path.

Python Code

```
graph = {  
    'A': [('B', 1), ('C', 3)],  
    'B': [('D', 3), ('E', 6)],  
    'C': [('F', 2)],  
    'D': [],  
    'E': [('G', 1)],  
    'F': [('G', 2)],  
    'G': []  
}
```

```
heuristic = {  
    'A': 6,  
    'B': 4,
```

```

'C': 4,
'D': 3,
'E': 1,
'F': 2,
'G': 0
}

open_list = [('A', 0)]
visited = []

while open_list:
    open_list.sort(key=lambda x: x[1] + heuristic[x[0]])
    node, cost = open_list.pop(0)

    if node not in visited:
        print(node, end=" ")
        visited.append(node)

        if node == 'G':
            break

        for neighbor, weight in graph[node]:
            open_list.append((neighbor, cost + weight))

```

Output

A B E G

```
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 6)],
    'C': [('F', 2)],
    'D': [],
    'E': [('G', 1)],
    'F': [('G', 2)],
    'G': []
}

heuristic = {
    'A': 6,
    'B': 4,
    'C': 4,
    'D': 3,
    'E': 1,
    'F': 2,
    'G': 0
}

open_list = [('A', 0)]
visited = []

while open_list:
    open_list.sort(key=lambda x: x[1] + heuristic[x[0]])
    node, cost = open_list.pop(0)

    if node not in visited:
        print(node, end=" ")
        visited.append(node)

        if node == 'G':
            break

        for neighbor, weight in graph[node]:
            open_list.append((neighbor, cost + weight))
```

Enter Input here

If your code takes input, add it in the above box before

Output

Status : Successfully executed

Time:
0.0200 secs

Memory:
8.996 Mb

Your Output

A B C D F G

Result

Thus, the A* Search Algorithm was successfully implemented and the goal node was reached using the shortest path.

These are **simple, easy-to-write lab programs** suitable for AI practical records and university lab exams.